Skilled Migration and the Evolution of Regional Economic Complexity in the European Union

Research Proposal for Economic Complexity

2025

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AI use declaration

While creating the research framework, a Large Language Model (ChatGPT) was employed to help in identifying the research gap in the existing literature. This was investigated further to affirm the validity of the topic, and to make sure it was not researched so far indeed. For writing the research proposal, the same tool was used to refine the text, correct language errors and check consistency.

1. Introduction

Over the last two decades, growing attention has been aimed at understanding why some regions within the European Union (EU) are able to transition into more sophisticated, productive, and knowledge-intensive economic structures while others remain locked in low-complexity activities. Economic complexity theory (Hidalgo & Hausmann, 2009; Hausmann et al., 2014) provides a framework for examining these differences. While originally applied on national level, the ECI framework can be increasingly relevant for diagnosing development at the sub-national scale, particularly within the diverse European context. Research utilizing ECI at sub-national levels confirms that higher complexity often correlates positively with regional economic growth and human development, though analyses have shown that this relationship can be heterogeneous or non-linear (Mewes & Broekel, 2020; Hamdi et al., 2025). For instance, complex regional economies, especially large cities, tend to attract both high- and low-skilled workers to support intricate organizational structures, potentially increasing competition and skill premiums (Hartmann & Pinheiro, 2022).

Human capital is an inherent component reflected in the ECI, meaning that skilled labor inputs are critical for boosting productive complexity. Tertiary-educated migrants represent a specific, highly mobile, and potent form of human capital that directly contributes to economic capabilities, also by being the flow of knowledge, skills, and technology between regions with different specializations (Caous & Huarng, 2021). This transfer mechanism might be vital for diffusing tacit knowledge and enabling regional economic diversification.

While previous research has examined how skilled migration affects innovation outputs and firm performance (e.g. Glennon, 2024), little is known about whether high-skilled migrants contribute to regional economic complexity upgrading, that is, whether they help regions move into more complex industries or production capabilities over time. Most studies analyzing ECI focus either on national-level determinants or on internal industrial structures, without considering migration-driven human capital inflows. Conversely, the migration literature rarely integrates the complexity perspective, even though this framework is well-suited to capturing the capability-based transformations that migrants may influence.

This research aims to bridge these literatures by empirically assessing the relationship between skilled migration and changes in regional economic complexity in the EU. Using a panel dataset

of NUTS 2 regions, the study aims to answer the question of "To what extent do skilled migrants contribute to increases in regional economic complexity in the European Union?"

2. Data

The empirical analysis will rely primarily on publicly available regional data from the Eurostat database. The unit of analysis is the nomenclature of territorial units of the European Union. NUTS 2 regions are defined as the basic regions for the framing of EU regional policies. These units typically encompass a population range of 800,000 to 3,000,000 (Eurostat, 2024). Furthermore, crucial economic and demographic data, including Business Enterprise Research and Development (BERD) statistics and detailed regional labor market indicators for migrants, are reliably compiled at the NUTS 2 level. The planned period of analysis is expected to cover approximately 2012-2024 and 200 regions, but it is subject to data completeness.

The dependent variable will be the change in the Economic Complexity index of the region (ΔECI). Because no official ECI exists for EU regions at the NUTS 2 level, regional complexity will be constructed manually using the method proposed by Hidalgo and Hausmann (2009). For the computations, industry employment data will be utilized, which is widely used as a proxy for capturing the underlying productive capabilities in the regional context, since trade data is often unavailable at sub-national scales. The procedure involves the following steps: first, we construct a bipartite region-industry matrix using Eurostat employment data disaggregated by NACE Rev. 2 codes, identifying whether a region has revealed comparative advantage in each industry. This binary matrix is then used to calculate the iterative measures of regional diversity and industry ubiquity that ultimately yield the ECI. The resulting eigenvector will be normalized to obtain a continuous measure of economic complexity for each region and year. Because the interest of the study lies in structural transformation rather than static levels, the analysis focuses on annual changes in ECI, that is:

$$\Delta \mathrm{ECI}_{r,t} = \mathrm{ECI}_{r,t} - \mathrm{ECI}_{r,t-1}$$

The key explanatory variable will be the share of tertiary-educated foreign-born residents in each region. This will be obtained using Eurostat migration and education statistics, which provide information on the population by country of birth and educational attainment at the NUTS 2 level. The variable will be lagged by one year to mitigate concerns about reverse causality. In our case, this means that regions that experience fast growth in complexity are simultaneously more attractive to skilled migrants, leading to a potential bias in the estimates.

Moreover, a set of regional-level controls will be included to account for alternative drivers of complexity growth, to isolate the effect of skilled migrants from broader economic, demographic, and human capital factors. All the variables are contained in the Eurostat database and they include:

- GDP per capita (Purchasing Power Standard),
- population density (persons per km^2),
- R&D expenditure per capita (million Euros per inhabitant at Purchasing Power Standard),
- share of natives in tertiary education (% of population).

3. Econometric Specification

3.1. The model

The core empirical model is a fixed-effects panel regression estimating the relationship between skilled migration and regional economic complexity growth. The baseline specification is the following:

$$\Delta \mathrm{ECI}_{r,t} = \alpha + \beta\, MIG_SKILL_{r,t-1} + \gamma\, X_{r,t-1} + \mu_r + \lambda_t + \varepsilon_{r,t}$$

Where:

- $\Delta \mathrm{ECI}_{r,t}$ is the change in economic complexity for region (r) between periods
- $MIG_SKILL_{r,t-1}$ is the lagged share of tertiary-educated migrants
- $X_{r,t-1}$ is the vector of control variables
- μ_r captures time-invariant regional characteristics
- λ_t captures year-specific shocks.

This specification directly addresses the research question by examining how the lagged share of skilled migrants affects the subsequent change in economic complexity. Lagging the migration variable addresses simultaneity concerns, as migration may be partly driven by pre-existing regional complexity. Also, economic complexity reflects structural changes in production capabilities, which do not respond immediately to new inflows of human capital. The effects of skilled migration on regional industries may operate through knowledge transfers and formations of new networks and business relationships. These processes take time to translate into measurable changes in ECI, therefore lagging the variables better captures the causal process over time.

The coefficient β captures the absolute change in ECI associated with a one-unit increase in the lagged share of tertiary-educated migrants. Keeping the skilled-migrant variable in its percentage form preserves a straightforward policy interpretation, as β directly reflects how much ECI changes when the share of skilled migrants rises by one percentage point.

The regional fixed effect (μ_r) controls for all unobserved, time-invariant regional characteristics that might influence the baseline rate of complexity growth, such as deep cultural factors or intrinsic geography. Since the dependent variable is already in differences, the remaining μ_r captures a region-specific linear trend (or average growth rate) in ECI.

The time fixed effect (λ_t) accounts for common, unobserved shocks that affect all EU regions equally (e.g., global trade crises, common EU regulatory changes), enhancing identification.

Robustness checks will also be carried out as a part of the analysis. They will include testing different lag structures (e.g., two-year lag) and using subsample analysis (e.g., core vs. peripheral EU regions).

3.2. Potential Challenges

The primary concern is reverse causality (endogeneity) because regions experiencing faster growth in complexity (high $\Delta \mathrm{ECI}_{r,t}$) are simultaneously more attractive to skilled migrants, leading to an upward bias in the estimated β coefficient. Although using a lagged predictor helps mitigate simultaneity bias, it might not fully resolve reverse causality.

Another concern would be omitted variable bias which arises from unobserved factors that simultaneously affect both skilled migration and changes in regional complexity. For example, regional governance quality, sector-specific industrial policies, historical innovation capacity, or local institutional strength may influence a region's ability to attract highly educated migrants while also shaping its trajectory of complexity upgrading. If these factors are not fully captured by the available controls or fixed effects, the estimated coefficient on skilled migration may be biased.

4. Expected Outcomes

Based on existing research on skilled migration, innovation, and regional capability formation, the baseline expectation is that the share of tertiary-educated migrants will have a positive and statistically significant effect on regional economic complexity growth. Regions attracting a larger proportion of highly skilled foreign-born residents are likely to experience greater improvements in ECI, consistent with the idea that mobile human capital supports knowledge diffusion, enhances the local skill base, and facilitates diversification into more sophisticated activities.

Regarding the control variables, several patterns are expected. GDP per capita is likely to exhibit a positive association with complexity growth, reflecting that more developed regions possess stronger institutional capacity, better innovative capabilities, and more diversified industries that enable transition toward higher value-added activities. Population density may also contribute positively, as densely populated regions tend to host agglomeration economies, richer labor market pooling, and more frequent knowledge spillovers which are all factors that support capability upgrading. Similarly, R&D expenditure per capita is expected to be one of the strongest predictors of rising complexity. Investment in research and development expands local technological capabilities, fosters innovation, and helps regions enter new, less ubiquitous industries, creating a direct channel through which complexity can increase.

Finally, the share of natives with tertiary education is anticipated to correlate positively with ΔECI, although likely more modestly than R&D expenditure. A highly educated native workforce provides a strong foundation of knowledge and skills that can be transformed into productive capabilities. Together, these expected outcomes would demonstrate that skilled migration contributes to (but does not exclusively drive) economic complexity upgrading. Instead, it interacts with other regional economic, demographic, and knowledge-related conditions, all of which shape the ability of regions to diversify and enhance their productive structures over time.

5. Impact

This research has substantial implications for EU regional policy, migration strategy, and smart specialization agendas. If skilled migration is shown to facilitate an increase in complexity, then policies aimed at attracting, retaining, and integrating highly educated migrants could become central tools in regional development strategies. Moreover, the findings would provide

empirical evidence supporting the idea that human capital flows are a source of knowledge and capability formation which are key components of the economic complexity framework.

The study also contributes to the field by integrating migration economics with complexity analysis, offering a new perspective on how regional economies accumulate and transform productive knowledge. By providing a rigorous empirical assessment at the regional level, the research enhances understanding of the mechanisms through which skills and knowledge diffuse across space.

6. Conclusion

This project investigates an important and underexplored question at the intersection of economic complexity and skilled migration. By constructing a panel dataset for EU NUTS 2 regions, computing regional ECI manually, and applying fixed-effects econometric methods, the research aims to identify whether skilled migrants contribute to structural economic development. The expected findings have direct policy relevance for innovation, regional development, and migration governance within the European Union. This study therefore represents a valuable contribution to both theory and evidence on the drivers of economic complexity in modern economies.

7. References

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